Math-M-Addicts
MINDS THAT COUNT

SP250 Homework 4

1. The rule for divisibility by 11 is as follows:

Consider any positive integer N. **Starting from the right**, compute the sum of all the digits in the even positions (call that sum E) and the sum of all the digits in the odd positions (call it O).

Then $N \equiv E - O \ (mod\ 11)$.

For example: N = 193257. E = 7 + 2 + 9 = 18. O = 5 + 3 + 1 = 9. According to the rule, N gives remainder 18 - 9 = 9 when divided by 11.

Prove this rule.

Solution:

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2. Prove, **using ONLY the material we have covered so far** and, in particular, no fancy factorizations, that for any two integers a and b with $a \neq b$ and positive integer n:

$$a - b \mid a^n - b^n.$$

Solution:

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3. Prove that from any 100 integers one can pick 15 such that the difference between any two of them is divisible by 7.

Solution:

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4. Is it possible to tile a 10x10 square with 25 1x4 rectangles? For the avoidance of doubt, tiling means that every one of 100 little squares belongs to exactly one 1x4 rectangle. Prove your answer.

Solution:

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5. Prove that equation $x^2 = 4y - 1$ does not have any integer solutions.

Solution:

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6. a) What is the maximum number of chess rooks that one can place on a chessboard so that no two rooks attack each other? Prove your answer.

b) What is the number of ways to place rooks in such a way?

Solution:
